Professional Issues

One of the most pressing issues in contemporary AI development is the environmental damage that excessive computational costs can impart on the environment. As I explored while training my ML agents, pursuing a “perfect” or “intelligent” model can be extremely computationally expensive. Pursuing a “perfect” model is governed by the Efficient Compute Frontier, a concept brought up in “Scaling Laws for Neural Language Models” by Kaplan et al. In which, Kaplan demonstrates that as a model reaches its optimal state, achieving marginal gains in model performance often requires an exponential increase in computation power.[1] This exponential requirement for computing power is often attributed to be the reason behind AI’s environment damaging reputation, due to the large carbon dioxide emissions[2] and excessive water consumption, especially in areas with limited fresh water available. [3]

From an ethical standpoint, any developer working with this new technology must acknowledge the externalities of their actions, as large AI related projects can quickly spiral out of control when related to the amount of compute power needed. In the case of my project, while training a ML agent through Reinforcement Learning to traverse a track may require hundreds, if not thousands of iterations running on a GPU, it may not have a sizable impact on the environment. However, a developer training a more advanced model may use significantly more power to run a reinforcement training program for more steps, using exponentially more power for a small improvement in model competence.

As such, all AI developers have a responsibility to lend significant thought towards algorithmic efficiency, to try reduce the footprint in training their models as opposed to taking the brute force approach with compute power that many large companies are taking.

In my project, I addressed this by ensuring a balance between model competency and compute power required. Once a model has shown itself to be able to complete the task reliably and with a stabilised mean reward, I decided to stop training in order to cut down on compute time as a small improvement to my models performance would likely not be noticed by the vast majority of end users.

Further related to AI, with an introduction of this new technology, especially to the software and game development sector, there is potential for job replacement, where many roles such as play testers or texture artists may be replaced by AI. This is especially present with beginner roles that are mostly comprised of tasks that AI is increasingly improving at, with some sources showing that roles which can be automated have had a sharp decline in job listings.[4] However, some sources point to new roles being created, especially in technology and data-related fields[5].

Professional bodies such as the Association for Computing Machinery (ACM), emphasise a commitment to reduce AI dependency and use it as a tool for augmentation, as opposed to replacing critically human aspects, such as critical thinking.[6] In my project, the role of ML agents were always going to be computer controlled, as no multiplayer features were within

the project's scope originally outlined. Therefore no human roles were automated during the development of this product.

Another problematic side effect of AI's prominence is the flooding of "slop" on the internet, in which often poorly researched, poorly made and low effort AI generated media is uploaded to the internet.[7] This often dilutes the significant human effort and craftsmanship associated with creating the same media in a traditional workflow, which can threaten the viability of the creative economy and the trustworthiness of most forms of media. With such a low barrier to entry to create media, the market can quickly be saturated by low quality content that can frustrate users and sap traffic away from human creators that are forced to put in more and more effort to compete[8], with a New York Times investigation showing that more than 40% of videos recommended to children can be categorised as "slop"[9]. This is further exemplified by a large proportion of internet traffic being bots, with some estimates of bot traffic being above 50% of all internet traffic.[10]

Due to the nature of my project being a racing game, there is the issue of vehicle brand licensing that affects many games, as licensed vehicles can heavily increase the desirability of a video game, as users want to drive cars they aspire to own in real life. However, this brings significant legal and ethical concerns regarding Copyright and Trademarks.

As vehicle manufacturers hold exclusive rights to their designs, branding, and other intellectual property, featuring these brands without a license is a legal risk and could be considered unethical as you are misleading the end user into thinking these vehicles are approved by the manufacturer.

Using the model I used for development (An Audi R8), was a technical choice as it was a model I had used before for personal use. This vehicle was also used as a baseline to calibrate my model with a vehicle I could reference in the real world, and coupled with the challenge of finding a car model that is not based on a licensed vehicle, I opted to use the Audi R8. However, This represents a form of Legal / Technical Debt,[34] while permissible for a personal / academic project with no intentions of a release.[11] If I were to want to release this project, I would have to go back and rework the vehicle in the game.

In conclusion, the development of this racing game has prompted me to think deeper about my actions as a developer, and the responsibilities I have in respect to the environment, legal system, and general ethics related to using the work of others in my own. Whether it be finding a balance between compute time and model performance, or the legal issues with using a 3d model of a licensed vehicle, this project has served as a large stepping stone towards my continued thought into the ethics of the software engineering career path I intend to enter.

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